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Yashasvi Baweja

Python Developer

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AI researcher turned quantitative software developer with hands-on experience in designing, enhancing & maintaining risk mitigation tools for fintech space

Education

Johns Hopkins University (JHU) GPA: 3.9/4.0 — Baltimore, MD
Master of Science (M.S) - *Electrical & Computer Engineering* — 2019 - 2022

Indraprastha Institute of Information Technology (IIITD) GPA: 8.31/10.0 — Delhi, India
Bachelor of Technology (B.Tech) - *Computer Science & Engineering* — 2015 - 2019

Skills

Programming Languages: Python, Bash, SQL, Java, C/C++
Cloud(AWS): Lambda(Serverless), Cloudwatch, Amplify, Cloudformation
Machine Learning: PyTorch, Tensorflow, Numpy, Pandas, OpenCV, Pillow, Jupyter, Scikit-learn
Frameworks: Excel, Dash, Flask, Django, Docker, SSH, Postman, VSCode, Debugger, Git
Databases: MySQL, Postgres, MongoDB, DynamoDB(NoSQL), AWS RDS, S3

Experience

Quant Developer / Risk Management — **May 2023 — Oct 2025**
Hull Street Energy, LLC — *Bethesda, MD*

- Built, automated, and maintained data feeds from external sources to reduce latency in work flows
  - Led development of pulling ***Platts*** America Gas & Power daily prices in internal database via Python ***APIs***
  - Built GUI symbol browsers in ***PyQt*** to map external codes with existing firm nomenclature
  - Maintained feeds from **commodities** derivatives exchanges (***ICE, CME***) and made data available to traders by ensuring proper data retrieval, cleaning & storage
  - Boosted data reliability by automating weekly data transfer using **Cron** to fill missing values
- Enhanced firm-wide risk assessment capabilities by building tools for portfolio & asset management
  - Structured and priced negotiation terms for an asset deal which resulted in **$1 MM/year** over 5-year period
  - Built ***Dash*** app to view booked trades in system enhancing transparency for optimal hedging
  - Added gas daily prices in Deal-Driver (modelling tool) to optimize for near term forecast
- Streamlined budget process to eliminate redundant methods & improve reporting efficiency
  - Worked with financial planning & analysis (FP&A) & asset management team every budget cycle **(6 months)** to refresh forward valuations for 5 year plan
  - Presented variance analysis reports to senior management, isolating key drivers of gross margin shifts
  - Generated hourly shaped forward pricing for all portfolios, directly supporting trading, hedging, and capital allocation decisions
- Managed the complete recruiting life cycle for strategists team hires (intern and full-time)
  - Streamlined interview logistics, coordinating internal schedules to conduct timely and effective assessments
  - Served as primary mentor for new hires, provided training on company processes & tools to ensure successful integration

Python Developer / Data Platforms — **Sep 2022 — Apr 2023**
CalaHealth Inc — *Baltimore (Remote)*

- Built a new microservice in patient portal which improved securing the registration over previous workflow
  - Deployed a serverless function in ***AWS Lambda*** to sync new user data from Salesforce in ***DynamoDB***
  - Utilized ***jwt token*** to verify signature in API to improve upon previous un-authenticated routine
- Created a REST API routine to assist research team in analysing logs
  - Spin up an ***AWS EC2*** instance to run a ***Flask*** app accepting logs from device base station(s)
  - Improved security of server by manually setting ***iptables*** to allow only tcp connections on specific port
- Ideated and developed RESTful microservices for the Health Care Personnel (HCP) dashboard
  - Demonstrated the importance of HCP portal by building individual microservices in ***Docker***
- Provided support to implement multiple APIs & fixing bugs reported in other microservices written in ***Python***

Researcher / Face Anti-Spoofing **Aug 2019 — May 2021**
Vision(ECE) Lab, JHU *Baltimore, MD*

- Addressed challenge of detecting spoofs in face authentication videos using anomaly detection
  - Proposed a training framework with only real face data while detecting spoofs as anomalies
  - Reduced latency in algorithm by selecting only one image per 30 frames using ***OpenCV***
  - Handled class imbalance in data by penalizing errors arising from under-sampled class
- *Published research at IJCB, 2020*
  - Received award for top 3 technical presentations

Projects

Chat GPT Demo
- Designed and implemented a search functionality using ChatGPT's API
  - Developed Django application which takes user query and displays response from ChatGPT API
  - Dockerized the entire application to remove system dependency
  - Served the application on AWS cloud using EC2 server. Currently live at: link

Movies Recommendation system
- Analysed Epinions data to build a recommendation system based on user choices
  - Modelled trust from social graphs in recommender systems to improve Mean Absoulte Error
  - Extended Node2Vec accepted at KDD'16 to learn continuous feature embeddings for nodes in networks
  - Built a Django app to allow users to rate movies upon entry

Restaurant Hygeine classifier
- Analysed 2021 data from Open Portal to find the best restaurants in San Francisco based on hygiene
  - Perform feature engineering on columns like business-location, zipcode, inspection score to find best features
  - Compared Random Forest, Decision Tree, Neural Network, k-NN & SVM classifier over 10-fold cross validation
- Gained insights that location alone is not a good predictor for hygiene

Voice2Face
- Using Generative Adversarial Networks to create emotionally aware faces from voices
  - Implemented the paper "*Reconstructing faces from voices, Wen et. al*" in PyTorch
  - Added new auxiliary network which handles emotion along with speech signals

Periocular Recognition
- Developed a novel loss for training CNNs in presence of data from different modalities
- Extended Triplet Loss by adding two branches(spectrum/resolution) to account for heterogeneity
- *Published research at BTAS, 2018*
  - Achieved State-of-the-art(10% improvement in EER) for periocular recognition

Face recognition on golf carts
- Led R&D team in building face recognition system (written in Python) for Yamaha Research(Japan)
- Built a personalised greeting tool using Haar cascade for segmentation and LightCNN model for classification
- Enhanced content information over video frames using multiple score fusion techniques

Activities & Awards

- Recipient of 2019 Graduate Student Fellowship, ECE department, JHU
- Student mentor for Intl Conference: CVPR, 2022 (New Orleans, Louisana)
- Project Showcase finalist (Smart Glasses), Mini-Maker Fair @ IIITD
- Reviewer for IEEE Transactions Journal on Image Processing, 2022
- Top 3 awardee for best presentation at IJCB, 2020

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